

State, Not Tokens: Repository-Scale Agent Reasoning Is Bound by State Architecture

Working draft — professorpalmer. Numbers marked [live] are still being collected; this draft reports only what the hardened oracle has actually verified.

Abstract

The agent community has largely treated repository-scale forgetting as a *context-window* problem: bigger windows (8k $\rightarrow$ 128k $\rightarrow$ 1M) are expected to yield better whole-repo reasoning. We argue this is a misdiagnosis. Using a hard, machine-checkable task — strict JavaScript $\rightarrow$ TypeScript migration of real OSS repositories under an unforgeable oracle (strict `tsc`, immutable test suites, mandatory `.js` $\rightarrow$ `.ts` replacement, zero type-escape-hatches) — we vary a single axis: **how state flows between bounded workers**. Three arms hold model, tools, scaffold, and oracle constant: a single-context *monolith*, a *durable* arm that accumulates each completed dependency layer as a committed artifact on a shared evolving tree, and a *stateless-RAG* arm whose per-file workers retrieve context but never see each other’s results. We find: (1) a single modern agentic worker already scales much further than the naive context thesis predicts — it cleanly migrates up to **240** interdependent jsdom modules by navigating the filesystem on demand rather than cramming a working set into the prompt — but it **does crack at the full 364-module tree**, leaving residual strict-type errors on the hardest module (a *capacity* failure, not a conversion failure); (2) when work is decomposed for parallelism, **durable accumulation strictly dominates stateless retrieval** — RAG’s independent workers emit code that does not even compile (TS2451 redeclaration conflicts appear *only* in RAG), while durable does not; and (3) durable state confers two structural properties no single transcript can: **interruption-resumable consistent checkpoints**, and **zero-marginal-cost re-query** of any materialized discovery (a SQLite read, not an LLM call). A failure taxonomy shows three architectures fail in three distinct ways — RAG by *conflict*, monolith by *capacity*, durable by neither. We also measure the limit of the parallelism this enables: the dependency critical path falls to **4.6% of total work** at full scale (so headroom *grows* with repo size), but *usable* concurrency is capped at **K $\approx$ 10–11** simultaneous agent sessions by the serving platform, not by durable state — an orchestration constraint we localize and leave as future work. The contribution is a reframing — *state is an asset, not a prompt* — with controls that isolate which capability actually matters.

1. Introduction

The dominant response to repository-scale forgetting in coding agents has been to enlarge the context window: 8k gave way to 128k gave way to 1M tokens, on the premise that whole-repo reasoning is gated by how much of the repository fits in the prompt at once (Liu et al., 2023; Packer et al., 2023). This paper argues that for repository-scale work the premise is a *misdiagnosis*. A modern agentic worker does not need to hold the working set in its prompt — it navigates the filesystem on demand, reading modules as it needs them — and as a result it scales much further than the naive context-length thesis predicts. Where it eventually fails, it fails by *capacity* (it cannot maintain global correctness over the hardest module), not by running out of window. The binding constraint is **how state is organized and flows**, not nominal context length.

The analogy is databases. Scaling was not solved by giving every process unlimited RAM; it was solved by making the process a *coordinator over durable state* — indexes, caches, query planners, materialized views — rather than the *container* of state. A query planner does not keep the whole table in memory; it reasons over durable structures and pulls what it needs. We claim repository-scale agents are undergoing the same transition: the model should be a navigator and reasoner over a durable, evolving world model of the repository, not a vessel that must hold the entire working set in a transient prompt and re-derive it every turn.

The reviewer’s sharpest question for any such claim is: *what does durable state buy that retrieval alone does not?* “Isn’t this just RAG over a code graph?” Our answer is mechanistic and measured, not rhetorical. We hold model, tools, scaffold, and a hardened oracle constant and vary a single axis — how state flows between bounded workers — across three arms: a single-context **monolith**, a **durable** arm that accumulates each

completed dependency layer as a committed artifact on a shared evolving tree, and a **stateless-RAG** arm whose workers retrieve context but never see each other’s results. RAG *finds* information and throws it away each turn; durable state *accumulates* discoveries, lets work decompose across workers without re-derivation, and makes prior findings consistent and reusable. Those are different capabilities, and the controlled design isolates them.

Contributions. 1. A controlled study design for repository-scale agents that varies *state architecture* with model/tools/oracle held constant, over an *unforgeable* oracle (strict `tsc`, immutable tests, mandatory `.js`→`.ts` replacement, zero escape hatches) and an independent variable of *repository scope*, not prompt tokens. 2. A refutation of the naive context-length thesis (the single context scales cleanly to 240 interdependent modules) and its reframing: the single context *does* crack at full-repo scale, but by capacity, not window — sharpening the contribution to “state architecture, not context length.” 3. A mechanistic account of *why* durable accumulation beats stateless retrieval: a failure taxonomy in which the three architectures fail three distinct ways — RAG by *conflict*, monolith by *capacity*, durable by neither — with TS2451 redeclaration conflicts appearing *only* in RAG. 4. Two structural properties unavailable to any single transcript: interruption- resumable consistent checkpoints, and zero-marginal-cost re-query of materialized discoveries (a database read, not an LLM call). 5. An honest localization of the one place durable does *not* win — wall-clock at full scale — to a *platform* concurrency ceiling ($K \approx 10$ –11 sessions), not to state architecture, with the parallel headroom shown to *grow* with repo size.

2. Related work

Long-context language models. A large body of work extends usable context length and studies its limits. “Lost-in-the-middle” effects show that simply enlarging the window does not yield uniform access to its contents (Liu et al., 2023), and context-management methods such as MemGPT page information between tiers to fit more useful signal into a bounded window (Packer et al., 2023). Our results are complementary but make a different point: for repository-scale coding the agent need not carry the working set in-window at all — MemGPT still treats the *window* as the substrate to be fed, whereas we treat durable committed state as the primary substrate and the window as incidental — so the relevant axis is state organization, not window size.

Retrieval-augmented generation and code-graph retrieval. RAG supplies models with relevant context on demand (Lewis et al., 2020), and repository-level methods such as RepoCoder iterate retrieval and generation over code (Zhang et al., 2023). We include a stateless-RAG arm with exactly this capability — per-file workers with code-graph retrieval — and show that retrieval alone is insufficient when work is decomposed: without shared, accumulated state, independent workers emit conflicting declarations that do not compile. The contribution is precisely to separate *retrieval* (finding context) from *accumulation* (persisting and reusing discoveries consistently).

Agent memory, scratchpads, and external stores. Prior work equips agents with memories and reflective stores to persist information across steps — e.g. Reflexion’s episodic memory of self-critiques (Shinn et al., 2023) and MemGPT’s OS-style memory paging (Packer et al., 2023). We treat durable state not as an auxiliary memory but as the *primary computational substrate*: discoveries are committed system objects on a shared evolving tree, which yields conflict-free decomposition, resumable checkpoints, and zero-cost re-query — properties we measure rather than assume.

Repository-scale agent benchmarks. SWE-bench evaluates agents on real GitHub issues (Jimenez et al., 2023). We differ on three counts: (i) the independent variable is *state architecture* with everything else held constant, not end-to-end agent capability; (ii) the oracle is *unforgeable by construction* (strict typecheck + immutable tests + mandatory file replacement + zero escape hatches), eliminating partial credit and hollow passes; and (iii) the scaling variable is *repository scope* drawn by deterministic BFS over the dependency graph, so we argue over repo size rather than token budgets.

3. Experimental design

3.1 Task and oracle

Strict JS→TS migration. Oracle PASS iff: `tsc --strict --noEmit` clean; immutable test suite green; **conversion-complete** (no in-scope `.js` left to shadow a `.ts` at runtime); zero escape hatches (**any**, **as any**, **@ts-ignore**, ... budget 0). The oracle propagates the tsx loader to spawned subprocesses so partial trees load `.ts` at runtime. Why this task: success is verifiable and *adversarially hard to fake* — partial credit and hidden hollow passes are eliminated by construction.

3.2 The single varied axis: state flow

- **monolith (C1)**: one worker, whole scope, one context.
- **durable (T)**: dependency-layer workers on a shared evolving tree; each inherits prior conversions (accumulation ON). Per-layer git commit = checkpoint.
- **stateless-RAG (R)**: per-file workers on pristine trees + code-graph retrieval, merged at the end (accumulation OFF; reuse only by re-derivation).

3.3 Independent variable: repository scope (not tokens)

Deterministic BFS over the intra-repo dependency graph from a fixed anchor, in size strata (jsdom S/M/L/XL/XXL/FULL = 8/24/60/120/240/364, where FULL is the entire `lib/` tree). We argue over repo size, not prompt tokens. Multiple seeds select different scope sets → generalization.

3.4 Metrics

Oracle pass; wall-clock; worker invocations; peak working set in one context; escape-hatch count; **Discovery Reuse Rate (DRR)** = fraction of dependent in-scope modules whose `.ts` consumes a type exported by an already-converted in-scope dependency (a persisted discovery); failure taxonomy.

4. Results

4.1 The naive context thesis fails at moderate scale — but the single context *does* crack at full-repo scale

scope (jsdom modules)	monolith	note
7 (express)	PASS	
8	PASS	
24	PASS	
60	PASS	0 hatches, 41 min
120	PASS	0 hatches, 46 min
240	PASS	0 hatches, 26 min
364 (full lib/)	FAIL	converts all, tests green, 0 hatches, but 16 strict-type errors

A single context did **not** break where the context-length thesis predicts: the agent is *already* a reasoner over external state (the filesystem), pulling in modules on demand rather than cramming a working set into the prompt. It cleanly migrated up to **240** interdependent modules. **This is itself a finding** and it reframes the contribution away from “state beats context for a single agent.”

But at the **full 364-module lib/ tree the one-shot architecture degrades**: it still converts everything and gets the test suite green with zero escape hatches, yet leaves **16 residual strict-type errors** (TS2322 assignability ×8, TS2571 “object is of type ‘unknown’” ×7), concentrated in the single hardest module (`XMLHttpRequest-impl`). Notably it failed *safely* — it reached for **unknown**, not **any** — but ran out of

capacity to globally narrow types at full scale. This is a **capacity** failure (correctness under a single global view), categorically different from RAG’s failure mode below.

Durable at the same 364 scale (capstone). Durable decomposition converted all 364 modules (364 workers, ~3.2 h) and its *raw* tree carries **only 5 type errors vs the monolith’s 16**. We are careful not to overclaim: durable does **not** eliminate conflicts outright. Its residue is bounded *intra-layer* — sibling modules converted in the **same** dependency layer are blind to each other (each sees only *prior* committed layers), so two CSS-rule siblings redeclared a shared type (TS2451/TS2300). This is the *same conflict class* RAG suffers, but **bounded to a single layer** instead of global. Because the tree is decomposed and consistent, **targeted iterative repair** — re-run only the conflicting modules, which can now see each other on the shared tree — drove the tree to a clean `typecheck_strict` in **3 iterations / 4 repair-worker calls / 258 s (~4.3 min)**, verified `tsc --strict --noEmit 0-error`. The one-shot monolith offers no such seam: its 16 errors sit inside a single 364-module context with nothing to re-run against. The durable edge at full scale is therefore **repairability via decomposition**, not a magically perfect first pass — a deliberately honest, narrower claim.

364-module full repo	monolith (one-shot)	durable (decomposed)
converted	364/364	364/364
raw strict-type errors	16	5
self-repair seam	none	per-module re-run
after targeted repair	n/a (no seam)	0 errors, CLEAN (4.3 min)

4.2 Durable accumulation > stateless retrieval (the clean divergence)

scope	durable (first pass)	durable (after targeted repair)	stateless-RAG
express (7)	PASS (DRR 0.75)	—	PASS (DRR 0.25)
jsdom-S (8)	PASS (seeds 1,2: 2/2)	—	FAIL (seeds 0,1,2: 0/3)
jsdom-M (24)	PASS (seeds 1,2: 2/2)	—	FAIL (seeds 1,2: 0/2)
jsdom-L (60)	localized gaps (s1: 1 err)	CLEAN (2 calls, 1.7 min)	FAIL (structural conflicts)
jsdom-XL (120)	CLEAN (typecheck, 0 hatches) ¹	—	—

Same scope, same model, same tools — only **accumulation** differs. The honest shape: at small scope (S, M) durable’s *first pass* is clean while RAG fails at **every** seed (S 0/3, M 0/2); at larger scope (L) durable’s first pass leaves a *few localized* type gaps that **targeted repair clears to CLEAN cheaply** (L seed 1: one TS2353, 2 repair calls, 101 s), whereas RAG fails by *structural cross-file conflicts* that have no cheap local fix. ¹At XL the durable runtime gate is confounded by jsdom’s own build system (§6); we report the unconfounded static axis (`tsc` clean, 0 hatches, all 120 converted).

The mechanism distinction is the point: RAG and durable both fail “by type errors” at scale, but **durable’s are localized and repairable** (a consequence of a consistent shared tree) while **RAG’s are structural conflicts** (a consequence of zero shared state). Accumulation does not just lower the error count — it changes the *kind* of error into one decomposition can cheaply repair.

4.3 Failure taxonomy: three architectures, three distinct failure modes

The mechanism is clearest in *how* each arm fails (TS error codes across all failing trials):

failure mode	arm	signature codes	reading
conflict	stateless-RAG	TS2451 redeclare $\times 10$, TS2717, TS2430, TS2739	blind workers emit colliding top-level declarations / inconsistent shared types $\rightarrow$ merged tree won't compile
capacity	monolith (364)	TS2322 $\times 8$, TS2571 un-narrowed unknown $\times 7$	global view, but can't maintain strict correctness on the hardest module at full scale
(none)	durable	—	shared state removes conflicts; decomposition bounds per-worker complexity

TS2451 (“cannot redeclare block-scoped variable”) appears **only** in the RAG arm and is the direct fingerprint of missing shared state. The monolith’s **unknown**/ assignability errors appear **only** under one global context at full scale. Durable avoids both by construction. This is the paper’s central mechanistic claim.

4.4 Discovery Reuse Rate

DRR cleanly separates accumulation from retrieve-and-forget where the codebase annotates with sibling types (express: durable/monolith 0.75 vs RAG 0.25). On jsdom’s CommonJS style DRR is a weaker, noisier discriminator; there the mechanism surfaces as the failure taxonomy (§4.3) rather than DRR. We report both honestly rather than cherry-picking the favorable metric.

4.5 Resumability (H4): the structural durable edge

At a hard interruption after an *equal* wall budget (~ 1200 s) on jsdom-M(24):

arm	work preserved @ crash	partial tree passes oracle?	recoverable on restart
durable	70.8% (17/24 committed)	layers type-check (consistent)	resumes $\rightarrow$ oracle PASS (24/24)
monolith	0% (0/24 committed)	no (inconsistent partial)	0 modules; full redo

Durable committed 17 modules across 6 dependency layers before the interrupt; resuming from that on-disk committed state completed the remaining 7 and the tree passed the full oracle. The monolith persists nothing until one terminal write, so its interrupted 24-file partial tree is inconsistent, fails the oracle, and yields zero known-good modules — the entire single shot is lost. This is a *structural* property of single-transcript execution, not a tuning artifact: there is no mechanism by which a monolith can expose a consistent intermediate checkpoint.

4.6 Hypothesis scorecard (calibrated, including refutations)

We pre-registered five hypotheses and report verdicts honestly — including where our own headline hypothesis was *refuted*, which sharpened the contribution.

H	claim (pre-registered)	verdict	evidence
H1	a single transcript’s pass-rate <i>collapses</i> once scope exceeds the window	REFUTED (naive form) → REFRAMED	monolith clean to 240 modules by navigating the filesystem; it does crack at 364 , but by <i>capacity</i> (un-narrowed types), not window-overflow. The binding constraint is state architecture, not nominal context length — the paper’s actual thesis.
H2	durable accumulation > stateless retrieval, same decomposition+retrieval	SUPPORTED	durable PASS vs RAG FAIL at S (0/3 seeds), M, L. Only accumulation differs.
H3	failure mechanism differs by architecture	SUPPORTED (revised)	not generic “context overflow”: RAG fails by <i>conflict</i> (TS2451 only in RAG), monolith by <i>capacity</i> (TS2571 only in monolith), durable by neither.
H4	durable resumes near-free under interruption; transcript must restart	SUPPORTED	durable 70.8% preserved + resume→oracle PASS; monolith 0% preserved, full redo.
H5	a larger-window model only postpones transcript collapse	NOT TESTED / MOOT	model held constant by design; the breaking-point analysis shows window was not the binding constraint, so the larger-window framing is superseded by H1’s reframing.

That H1’s naive form failed is the most important honesty in this paper: it is *why* the contribution is “state architecture, not context length,” not “we beat the context window.” The win is concentrated exactly where a single context is structurally insufficient — parallel decomposition without conflicts, and resumable consistent checkpoints — not in out-muscling a navigating agent at moderate scale.

4.7 Re-query cost: zero LLM calls (the asset property, made literal)

The defining property of an *asset* versus a *prompt* is that reuse cost $\rightarrow 0$. Durable state has it exactly: once a discovery is **materialized as an artifact**, recalling it is a database read — **zero LLM invocations**. On a completed conversion job, the full structured result (gate verdict, changed files, strict-typecheck outcome, and the provenance that it reused a sibling’s exported types) recalls in **<0.5 s with no model call**. We report the cross-arm re-query cost in **invocations**, not tokens — sidestepping the unreliable Cursor-SDK token counts with a categorical metric:

arm	cost to re-query a prior discovery	why
durable	0 invocations (artifact read)	discoveries are durable system objects
transcript	retain in-context (window-capped → fails at scale) or ≥ 1 (re-derive)	discoveries are transient text
stateless-RAG	≥ 1 invocation every time	re-retrieve + re-reason; nothing accumulates

DRR (§4.4) is the *empirical rate* of these zero-cost reuses: at the 364-module capstone, **86/293 dependency-bearing modules (29%)** consumed a prior worker’s materialized artifact. **Honest boundary:** “zero” is for *recall* of an already-materialized result; a re-query needing genuinely new synthesis still pays the synthesis — but it starts from the artifact (no re-derivation of the base), strictly cheaper than transcript or RAG. For *this* task the materialized discovery is a `.ts` file on disk, so a navigating agent can also re-read it cheaply (§5); the property is most decisive for **non-code reasoning artifacts** (analyses, decisions, traces) that do not live in the tree — flagged as the highest-value generalization.

4.8 Parallel headroom grows with scale, but usable concurrency is platform-capped

Decomposition exposes parallelism, and the *available* parallelism **grows with repo size**: the dependency critical path (longest chain that must run serially) shrinks as a fraction of total work as the DAG broadens (max layer width 5 → 78 across the sweep):

scope	critical path / total work	dataflow speedup headroom
8	45%	2.2×
24	21%	4.8×
60	20%	4.9×
364	4.6%	21.6×

So ~95% of full-scale work is parallelizable (Fig. `headroom_vs_scale`). But a soak that raised requested concurrency on the *same* full-scale durable run found a hard limit that is **not** durable state’s: worker success follows $\min(1, K/C)$ with **$K \approx 10\text{--}11$** (clean single runs: $C=16 \rightarrow 70\%$ [44/63], $C=32 \rightarrow 34\%$ [51/152]; Fig. `concurrency_ceiling`). Beyond ~K concurrent sessions the serving platform throttles excess workers into fast (<20 s vs ~90–170 s) no-edit returns. We attribute this to the **Cursor API/SDK session cap, not the orchestrator**: Puppetmaster spawned, leased, and retried all 32 correctly, and durable state + retry *absorbed* the throttle (the run still converges). The consequence: at the practical ceiling, durable wall $\approx$ work/10.6 $\approx$ 1.1 h vs the monolith’s 0.83 h — durable is $\sim 1.3\times$ *slower* on wall-clock while being the only arm that reaches a clean strict typecheck at full scale. The 21.6× theoretical headroom is real but only ~K of it is spendable at once today; closing that gap is an *orchestration* problem (§6 future work), not a property of durable state.

5. Discussion

- **Three advantages, one root.** Everything durable wins flows from a single property — *discoveries are durable system objects, not disposable prompt text*: (1) conflict-free parallel decomposition (§4.2–4.3); (2) interruption-resumable consistent checkpoints (§4.5); (3) zero-marginal-cost re-query of materialized discoveries (§4.7).
- What durable state does **not** buy for *this* task: raw single-agent navigation is already strong at moderate scale, and because the migration artifact is code on disk, re-reading a prior discovery is filesystem-cheap for any navigating agent too — so the zero-cost-re-query advantage is *under-tested* here and is most decisive for non-code reasoning artifacts. We do not claim otherwise.

- Implication: the durable advantage is an *orchestration/coordination* property, realized when one context is insufficient, or work must survive interruption, parallelize, or be revisited cheaply — not a universal “context is solved” claim.
- **The speed gap is an orchestration ceiling, not a state-architecture one.** Durable’s $\sim 1.3\times$ wall-clock penalty at full scale (§4.8) is set by a platform session cap ($K \approx 10\text{--}11$), not by accumulation; the theoretical $21.6\times$ headroom says the *architecture* scales, and realizing it is an engineering problem on the serving/scheduling side (§6).

6. Threats to validity

- **jsdom build-system coupling (scope-correlated).** jsdom’s `npm run prepare` runs `scripts/webidl/convert.js`, which `require()`s `lib/jsdom/living/helpers/namespaces.js` by *literal .js path*. On any trial whose scope includes that one helper, converting it to `.ts` makes jsdom’s own IDL codegen fail, which fails the *runtime* gate (`tests_api`). We verified this is *exactly* scope-membership-correlated (L-durable-s2 with `namespaces.js` $\in$ scope hit it; L-durable-s1 without it did not, same L(60) scale) — a target build artifact, arm-independent, not a durable-arm defect and not a scale effect. We therefore make affected comparisons on the **static** axis (`typecheck_strict` + `escape_hatches`), which jsdom’s build cannot perturb, and flag `tests_api` as confounded on those trials. A codegen-baked harness (run `prepare` on the pristine tree pre-conversion, exclude `scripts/` from scope) removes the confound; logged as future work rather than silently dropped.
- One task family (migration); artifact == code. Generalization to reasoning-artifact tasks is future work.
- Oracle hardening history (hollow passes, subprocess `.ts` loading, over-broad hatch counting) — all trees re-scored by the final oracle for internal consistency; the hatch fix flipped mono-240 from a false FAIL to its true PASS.
- Cursor-SDK token counts are unreliable (implausibly low); we report wall-clock, worker count, and DRR as the cost axes instead of token deltas.
- Single platform (Puppetmaster cursor workers); model routing held constant.
- **Platform concurrency ceiling (§4.8).** Usable parallelism is capped at $K \approx 10\text{--}11$ concurrent agent sessions by the serving API, so the measured wall-clock does not yet realize the $21.6\times$ dataflow headroom. This bounds the *speed* comparison (durable $\sim 1.3\times$ the monolith at full scale) but not the *correctness*, *resumability*, or *re-query* results, none of which depend on concurrency. We probed clean single runs at $C \in \{4, 16, 32\}$ to isolate the cap; a wider sweep and a second serving backend are future work.

6.1 Future work (orchestration track — distinct from the state-architecture claim)

These close the §4.8 speed gap and are properties of the *scheduler/serving layer*, not of durable state; we list them so the speed result is not mistaken for a ceiling on the architecture: - **Adaptive admission control.** Cap dispatch near the observed session ceiling instead of launching `max_workers` that mostly fast-fail; treat a burst of sub-N-second no-edit returns as backpressure and throttle, eliminating wasted API calls beyond K . - **Throttle-vs-failure classification.** A sub-N-second `require_diff` failure should re-queue as backpressure, not count against a task’s quality budget (no retry storms). - **Dataflow scheduling.** Release a module the instant its dependencies commit, rather than at full-layer barriers — the theoretical critical-path floor is $0.55\text{ h} < \text{the monolith’s } 0.83\text{ h}$, so dataflow + a higher session allowance is the path to a *speed* win, not just parity. - **Non-code reasoning-artifact tasks** to test the zero-cost re-query advantage (§4.7) where the discovery does not live on the filesystem.

7. Conclusion

Repository-scale agent performance is primarily constrained by **state architecture**, not nominal context length. A single modern agentic context scales much further than the context-length arms race assumes (clean to 240 modules by navigating the filesystem), but it does eventually crack at full-repo scale — and it

cracks by *capacity* (un-narrowed types under one global view), not by running out of window. When work is decomposed for parallelism or resilience, *how state flows between workers* becomes decisive: stateless retrieval fails by *conflict* (colliding declarations its blind workers cannot reconcile), while durable accumulation fails by neither, additionally buying interruption-resumable consistent checkpoints. The win comes from treating discoveries as durable, consistent, reusable system objects — *state is an asset, not a prompt*.

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